

unleash a virus attack to erase your entire system, steal confidential information from your system or even break into your system and modify your files without your knowledge.

A computer security professional does a similar job, albeit ethically. He thinks like a hacker so as to find loopholes in the data and systems in the network and pre-empt a hacker's moves. He then puts in place a security program to deal with the myriad of threats present for any infrastructure. There are many courses available nowadays that offer courses in "ethical hacking".

Computer Forensics is still a rather new discipline in computer security and is a rapidly growing discipline that focuses on finding digital evidence after a computer security incident (breach) has occurred.

The role of a Computer Forensic professional is to collect and analyse digital data within an investigative process and find out exactly what happened on a digital system and who was responsible.

There are essentially four phases for recovering evidence from a computer system or storage medium.

- The identification of digital evidence
- The preservation of digital evidence for judicial scrutiny in a court of law
- The analysis of digital evidence – the extraction, processing and interpretation of digital data
- The presentation of evidence in a court of law

Hundred per cent data security is a myth. The rapid evolution of computer technology implies that computer security professionals, despite being on constant vigil, may not be able to anticipate and prevent all attacks.

However, it is said that computer criminals always leave tracks; it's only a matter of finding them. But this part is not always easy. If computer forensics collects evidence correctly, it is much easier to apprehend the attacker and the evidence stands a greater chance of being admissible in the event of a prosecution.

Some organisations have put in place an 'Incident Response Team' whose role is to investigate in case of a computer security breach; this also involves problem-solving. It includes management personnel with the authority to take action, technical personnel (computer security professional and/or computer forensic professional) with the knowledge and expertise to diagnose rapidly and resolve problems, and communications representatives who can keep the concerned individuals and organisations informed on the status of the problem and develop public image control strategies, if necessary.

The Growth Story

According to a NASSCOM estimate, the world-wide demand for computer security professionals would touch about 2,00,000 by next year. In India alone, the demand should touch a whopping 90,000 and there would be a shortfall of 35,000 to 45,000 computer security professionals.

International Data Corporation (IDC) had predicted that the worldwide information security market would increase from roughly \$6.7 billion in 2000 to \$21 billion by 2005. In the financial services industry alone, the spending on security-related products and services was expected to rise from \$848 million in 2000 to \$2.2 billion by 2005. Thus, there is a sizable requirement for cyber security products and services.

Face-2-Face with practitioner

"My profession is IT Security. It is more a profession than a business. Yes, it is representative of career in IT security and forensics. In my profession, my job is to secure the IT assets of the end-user."

Rakesh Goyal
Sysman India